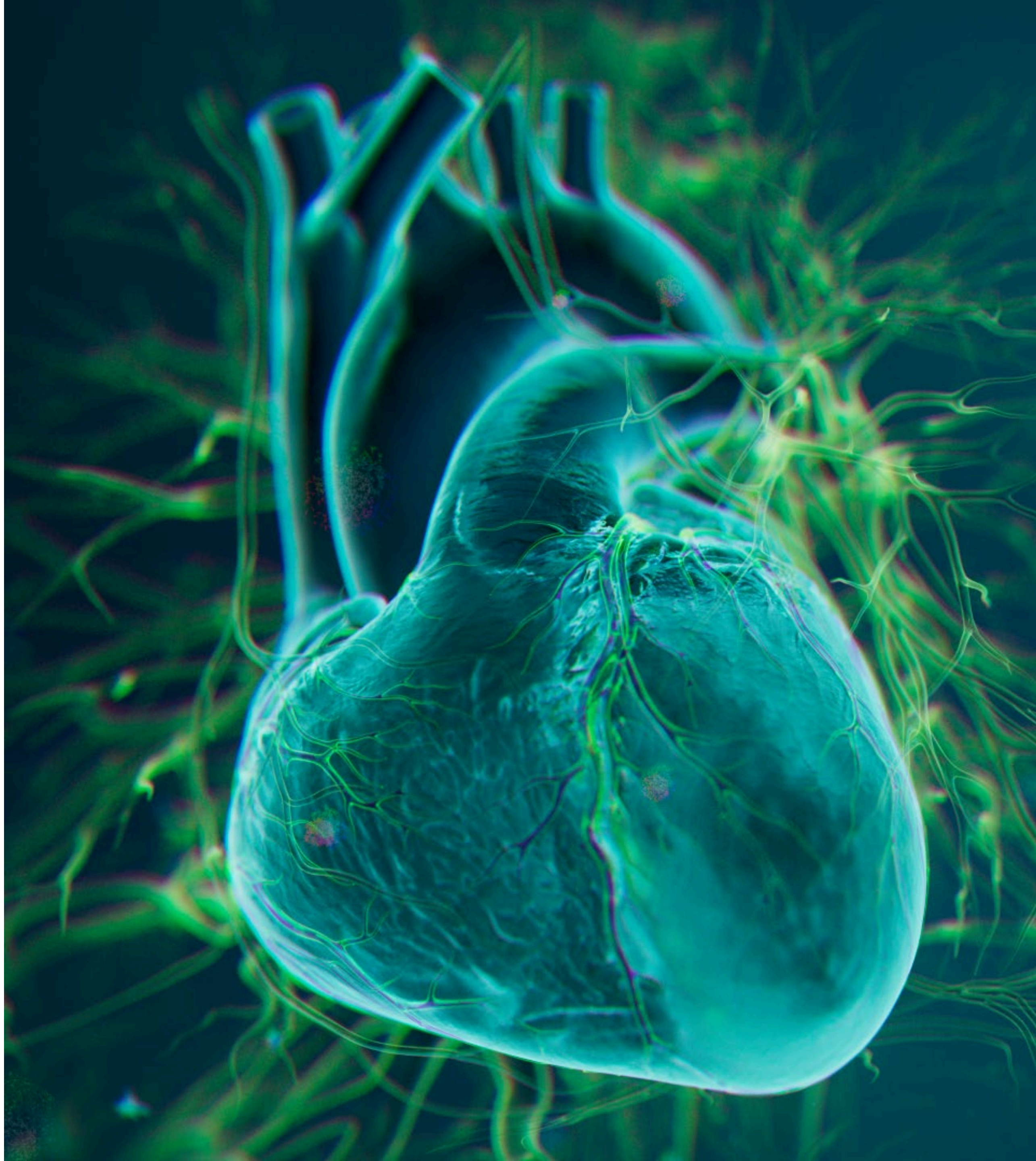
# American Heart Association 2025 Investor Presentation

November 9, 2025



# Forward-looking statement of Merck & Co., Inc., Rahway, N.J., USA

This presentation of Merck & Co., Inc., Rahway, N.J., USA (the “company”) includes “forward-looking statements” within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. These statements are based upon the current beliefs and expectations of the company’s management and are subject to significant risks and uncertainties. There can be no guarantees with respect to pipeline candidates that the candidates will receive the necessary regulatory approvals or that they will prove to be commercially successful. If underlying assumptions prove inaccurate or risks or uncertainties materialize, actual results may differ materially from those set forth in the forward-looking statements.

Risks and uncertainties include but are not limited to, general industry conditions and competition; general economic factors, including interest rate and currency exchange rate fluctuations; the impact of pharmaceutical industry regulation and health care legislation in the United States and internationally; global trends toward health care cost containment; technological advances, new products and patents attained by competitors; challenges inherent in new product development, including obtaining regulatory approval; the company’s ability to accurately predict future market conditions; manufacturing difficulties or delays; financial instability of international economies and sovereign risk; dependence on the effectiveness of the company’s patents and other protections for innovative products; and the exposure to litigation, including patent litigation, and/or regulatory actions.

The company undertakes no obligation to publicly update any forward-looking statement, whether as a result of new information, future events or otherwise. Additional factors that could cause results to differ materially from those described in the forward-looking statements can be found in the company’s Annual Report on Form 10-K for the year ended December 31, 2024 and the company’s other filings with the Securities and Exchange Commission (SEC) available at the SEC’s Internet site (www.sec.gov).

# Agenda

## Cardiometabolic and Respiratory Portfolio

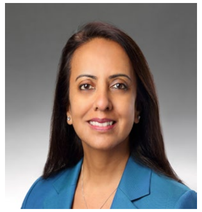
Dr. Dean Li
President, Merck Research Laboratories

## AHA Data and Enlicitide Strategy

Dr. Joerg Koglin
SVP, Head of General & Specialty Medicine, Global Clinical Development

## Addressing the Silent CV Epidemic

Jannie Oosthuizen
President, U.S. Human Health



## Q&A

Including Kshama Roberts
SVP, Global Pharmaceuticals

# Cardiometabolic and Respiratory Portfolio

Dr. Dean Li
President, Merck Research Laboratories

MERCK

# Delivering and developing therapies with the goal of redefining care in cardiometabolic and respiratory

★

## WINREVAIR®

**First and only** activin signaling inhibitor
Targets underlying disease pathophysiology

**Three positive Phase 3 trials:**
STELLAR
ZENITH
HYPERION

**Phase 2 CADENCE study in CpC-PH due to HFpEF**

**Autoinjector in development**

## Ohtuvayre

**First-in-class** selective dual inhibitor of PDE3 and PDE4 combining bronchodilatory and non-steroidal anti-inflammatory effects

**Launched in U.S. based on pivotal ENHANCE trial**

**Phase 2 study in non-cystic fibrosis bronchiectasis**

★

## Enlicitide

**Potential first-of-its-kind oral** PCSK9 inhibitor designed to deliver antibody-like efficacy in form of a pill

**Three positive pivotal trials:**
CORALreef Lipids
CORALreef HeFH
CORALreef AddOn

**Filings beginning in early 2026**

**Potential anchor for combinations**

★ Data presentations at AHA

# WINREVAIR continues to generate clinical evidence that reinforces potential to transform treatment of PAH

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Sotatercept for the Treatment of Pulmonary Arterial Hypertension

Marc Humbert, M.D., Ph.D., Vallerie McLaughlin, M.D., J. Simon R. Gibbs, M.D., Mardi Gomberg-Maitland, M.D., Marius M. Hoeper, M.D., Ioana R. Preston, M.D., Rogerio Souza, M.D., Ph.D., Aaron Waxman, M.D., Ph.D., Pilar Escribano Subias, M.D., Ph.D., Jeremy Feldman, M.D., Gisela Meyer, M.D., David Montani, M.D., Ph.D., Karen M. Olsson, M.D., Solaiappan Manimaran, Ph.D., Jennifer Barnes, Ph.D., Peter G. Linde, M.D., Janethe de Oliveira Pena, M.D., Ph.D., and David B. Badesch, M.D., for the PULSAR Trial Investigators*

**PULSAR Phase 2b study**
providing proof-of-concept

2021

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Phase 3 Trial of Sotatercept for Treatment of Pulmonary Arterial Hypertension

M.M. Hoeper, D.B. Badesch, H.A. Ghofrani, J.S.R. Gibbs, M. Gomberg-Maitland, V.V. McLaughlin, I.R. Preston, R. Souza, A.B. Waxman, E. Grünig, G. Kopeć, G. Meyer, K.M. Olsson, S. Rosenkranz, Y. Xu, B. Miller, M. Fowler, J. Butler, J. Koglin, J. de Oliveira Pena, and M. Humbert, for the STELLAR Trial Investigators*

**STELLAR Phase 3 study**
establishing risk benefit in broad population; served as basis for approval

2023

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 | MAY 29, 2025 | VOL. 392 NO. 20

Sotatercept in Patients with Pulmonary Arterial Hypertension at High Risk for Death

Marc Humbert, M.D., Ph.D.,[1] Vallerie V. McLaughlin, M.D.,[2] David B. Badesch, M.D.,[3] H. Ardeschir Ghofrani, M.D.,[4] J. Simon R. Gibbs, M.D.,[5] Mardi Gomberg-Maitland, M.D.,[6] Ioana R. Preston, M.D.,[7] Rogerio Souza, M.D., Ph.D.,[8] Aaron B. Waxman, M.D., Ph.D.,[9] Victor M. Moles, M.D.,[2] Laurent Savale, M.D., Ph.D.,[1] Carmine Dario Vizza, M.D.,[10] Stephan Rosenkranz, M.D.,[11] Yaru Shi, Ph.D.,[12] Barry Miller, M.S.,[12] Harald S. Mackenzie, M.B., Ch.B.,[12] Samuel S. Kim, Ph.D.,[12] Maria José Loureiro, M.D.,[12] Mahesh J. Patel, M.D.,[12] Joerg Koglin, M.D., Ph.D.,[12] Alexandra G. Cornell, M.D.,[12] and Marius M. Hoeper, M.D.,[13] for the ZENITH Trial Investigators*

**ZENITH Phase 3 study**
demonstrating reduction in risk of major morbidity and mortality outcomes and PAH hospitalization in advanced patients

2025

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Sotatercept for Pulmonary Arterial Hypertension within the First Year after Diagnosis

V.V. McLaughlin,[1] M.M. Hoeper,[2] D.B. Badesch,[3] H.A. Ghofrani,[4] J.S.R. Gibbs,[5] M. Gomberg-Maitland,[6] I.R. Preston,[7] R. Souza,[8] A.B. Waxman,[9] G. Kopeć,[10,11] G. Meyer,[12] K.M. Olsson,[2] W. Fu,[13] Y. Shi,[13] B. Miller,[13] S.S. Kim,[13] H.S. Mackenzie,[13] M. Brambatti,[13] M.J. Patel,[13] J. Koglin,[13] A.G. Cornell,[13] and M. Humbert,[14] for the HYPERION Trial Investigators*

**HYPERION Phase 3 study**
expanding benefit of reduction in risk of clinical worsening to recently diagnosed patients

2025

Each of the trials studied WINREVAIR on top of background PAH therapy versus background PAH therapy alone

# Data at AHA and ZENITH label update highlight profound effect on risk of major morbidity and mortality events

## Pooled post-hoc analysis

PULSAR, STELLAR and ZENITH studies show effect of WINREVAIR on **major morbidity** and **mortality outcomes** in **broad range** of adult patients with PAH[1]

| 75% | 56% | 51% |
|---|---|---|
| reduction in risk of composite morbidity and mortality endpoint | reduction in risk of lung transplantation or death | reduction in risk of death |

## ZENITH Label Update

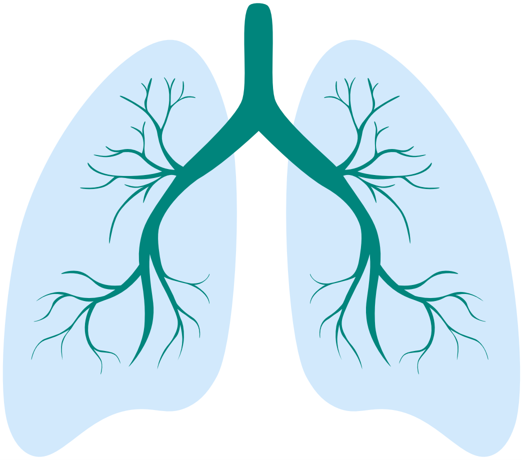
WINREVAIR™
(sotatercept-csrk) for injection 45 mg, 60 mg

Now the **first PAH therapy** with an indication that includes components of clinical worsening events: **hospitalization** for PAH, lung **transplantation** and **death**[2]

1. Data available in Appendix 2. WINREVAIR is indicated for the treatment of adults with PAH to improve exercise capacity and WHO functional class, and reduce the risk of clinical worsening events including hospitalization for PAH, lung transplantation and death.

# OHTUVAYRE is a first-in-class novel mechanism with broad label for maintenance treatment of COPD in adults

Ohtuvayre™
(ensifentrine) Inhalation Suspension
3 mg/2.5 mL



First novel inhaled mechanism for treatment of COPD in more than 20 years

Combines bronchodilatory and non-steroidal anti-inflammatory effects

Broad label, positive risk-benefit profile

Positive real-world experience continues to build

Opportunity for additional innovation and follow-on indications

**Potential to help address COPD**, a progressive disease and 4th leading cause of mortality globally

# Enlicitide is the first oral PCSK9i designed to have antibody-like efficacy

## **Profound efficacy** with **placebo-like safety** and tolerability now demonstrated in **two Phase 3 trials**

| CORALreef Lipids | | | CORALreef HeFH | | |
|---|---|---|---|---|---|
| LDL-C -59.7%[1] | | | LDL-C -59.4% | | |
| ApoB -50.3% | Non-HDL-C -53.4% | Lp(a) -28.2% | ApoB -49.1% | Non-HDL-C -53.0% | Lp(a) -27.5% |

**Achieved LDL-C reduction surpassing** other oral lipid lowering therapies used as add-on to statins

**Broad atherogenic particle reductions**[2] across non-HDL-C, ApoB, and Lp(a)

**Robust LDL-C goal attainment**[3]**:** Over two-thirds of participants achieved rigorous prespecified goal requiring both LDL-C <55 mg/dL and ≥50% reduction from baseline

**Ease of use:** safety comparable to placebo; no adverse DDIs, >97% medication compliance in CORALreef Lipids

1. Result from post-hoc reanalysis conducted to remove imputed biologically impossible baseline values. In primary analysis LDL-C reduction at Week 24 was 55.8% 2. Multiplicity-controlled secondary endpoints 3. Non-multiplicity-controlled secondary endpoint

# AHA 2025 Data Highlights: enlicitide decanoate

Dr. Joerg Koglin
SVP, Head of General & Specialty Medicine,
Global Clinical Development

# Building on the legacy that changed cardiovascular medicine

## Legacy of innovation in atherosclerosis

ZOCOR® (SIMVASTATIN)

MEVACOR® (LOVASTATIN)

Zetia® (ezetimibe) 10 mg Tablets

VYTORIN® (ezetimibe/simvastatin) tablets


**2025 American Chemical Society National Historic Chemical Landmark** recipient for development and commercialization of **statins**

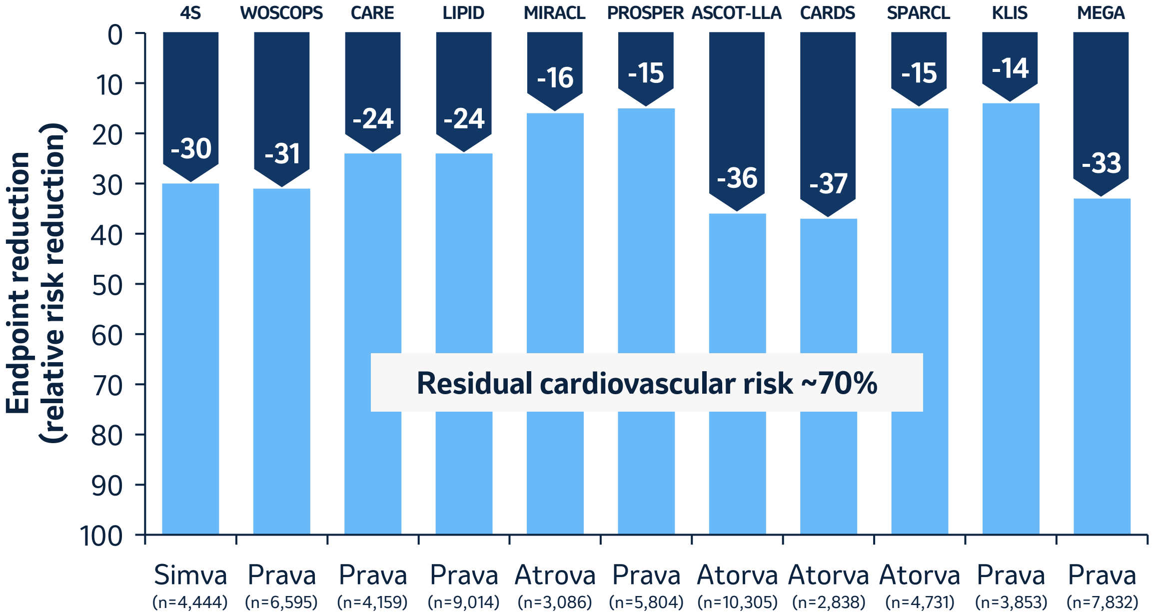
# Global CV epidemic claims 19 million lives per year

## Atherosclerosis


of all cardiovascular disease deaths attributed to ASCVD[1]

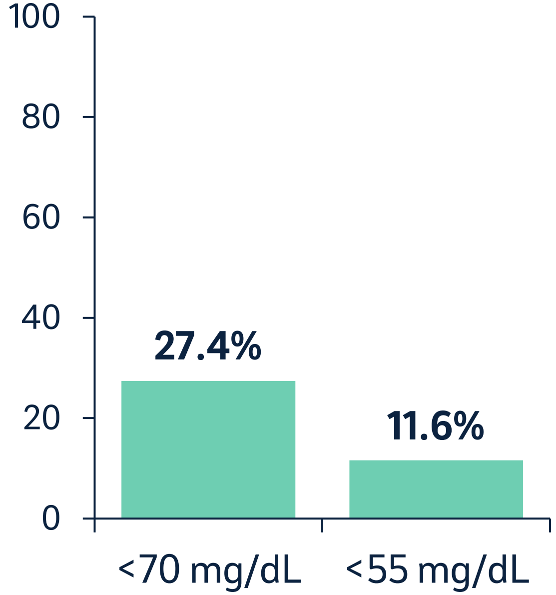
Elevated LDL-C is a risk factor for ASCVD

## Residual cardiovascular risk




Even with current treatment options, **residual CV risk remains high** if LDL-C is elevated[2]

## Patients at LDL-C goal




**<30%** of treated patients with ASCVD in the U.S. achieve guideline recommended LDL-C reductions[3]

1. World Health Organization. Cardiovascular diseases (CVDs). https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds). [Last accessed October 2021] 2. Chart derived from https://www.sciencedirect.com/science/article/pii/S0140673605673941?via% 3. Gu et.al., 2022 (analysis of 2019 Truven data) American Journal of Preventive Cardiology; Cannon et al

# Goal to deliver the most effective LDL-C lowering pill and make it accessible to a broad range of patients globally

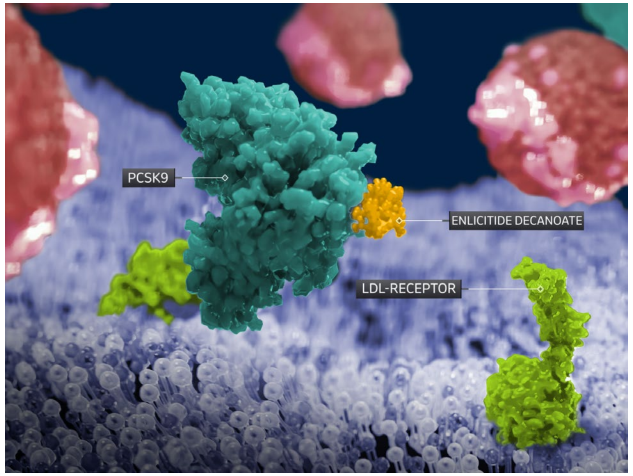
## PCSK9 monoclonal antibodies

### Direct PCSK9 binding

- PCSK9i mAbs act extracellularly and bind the PCSK9 protein itself in circulation, blocking its interaction with LDL receptor (LDLR) and preserving LDLR recycling and LDL clearance

Two PCSK9i mAbs on the market show **~50–60% LDL-C reduction with significant MACE reductions in large trials** and **favorable safety profile** but use has been limited to date

## Enlicitide




**Novel macrocyclic peptide platform** offers mAb-like selectivity at 1% the size

Shown to disrupt the binding of PCSK9 to LDLR in vitro in a similar manner as marketed mAbs but **administered orally once daily**

Ability to **manufacture and distribute at scale**

Aspiration to achieve **broad global access**

# Development program evaluating enlicitide across broad range of patients with ASCVD

| | CORALreef Lipids | CORALreef HeFH | CORALreef AddOn | CORALreef Outcomes |
|---|---|---|---|---|
| **Population** | Secondary prevention and intermediate to high-risk primary prevention patients (n=2,912) | Heterozygous familial hypercholesterolemia (n=303) | Secondary prevention and intermediate to high-risk primary prevention patients (n=301) | Secondary and high-risk primary prevention (n=14,550) |
| **Stable background therapy** | Moderate- or high-intensity statin (or statin intolerance) ± nonstatin lipid lowering therapy | Moderate- or high-intensity statin ± nonstatin lipid lowering therapy | Statin ± nonstatin lipid lowering therapy | Moderate- or high-intensity statin ± nonstatin lipid lowering therapy |
| **Comparator** | placebo | placebo | ezetimibe, bempedoic acid or ezetimibe + bempedoic acid | placebo |
| **Primary Endpoint** | Week 24 LDL-C change | Week 24 LDL-C change | Week 8 LDL-C change | MACE-Plus[1] |
| **Status** | ✓ Presented at AHA | ✓ Presented at AHA | ✓ Completed, To be Presented | Enrollment Complete Primary Completion Date November 2029 |

1. CHD Death-Based MACE-Plus is defined as any of the following: coronary heart disease death, myocardial infarction (MI), ischemic stroke (fatal and nonfatal), acute limb ischemia or major amputation, or urgent arterial revascularization (coronary, cerebrovascular, or peripheral).

# CORALreef Lipids provides evidence for potential first approval of an oral PCSK9i

**Inclusion Criteria:**

**Prior ASCVD event** with LDL-C ≥55 mg/dL
or
**at risk for first event** with LDL-C ≥70 mg/dL

**Treatment period: 52 Weeks**

R: 2:1

**Total randomized: 2,912**

Enlicitide 20 mg oral daily*

Placebo*

**8 weeks**

Follow-up or open-label extension

**Primary Endpoint**

- LDL-C mean percent change from baseline at Week 24
- Proportion of participants with AE; proportion of participants with discontinuations due to AEs

**Secondary Endpoints**

**Multiplicity-controlled secondary endpoints**

- LDL-C mean percent change from baseline at Week 52
- non-HDL-C mean percent change from baseline at Week 24
- ApoB mean percent change from baseline at Week 24
- Lp(a) median percent change from baseline at Week 24

**Other secondary endpoints**

- Proportion of participants with LDL-C <70 mg/dL and ≥50% reduction from baseline at Week 24
- Proportion of participants with LDL-C <55 mg/dL and ≥50% reduction from baseline at Week 24

*Participants must be on **optimized and stable background LLT with moderate- or high-intensity statin** (or with documented statin intolerance) ± other lipid lowering therapies

CORALreef Lipids

# Broad primary and secondary prevention study population

| Select study population characteristics | Enlicitide 20 mg n=1,940 | Placebo n=969 | Total n=2,909 |
|---|---|---|---|
| **Sex**, n (%) Female | 775 (40) | 367 (38) | 1,142 (39) |
| **Age**, mean (SD) years | 62.8 (11) | 62.7 (11) | 62.8 (11) |
| **BMI**, mean (SD) $kg/m^2$ | 29.2 (6) | 29.3 (6) | 29.2(6) |
| **History of major ASCVD event**, n(%) | 1,125 (58) | 572 (59) | 1,697 (58) |
| **Heterozygous Familial Hypercholesterolemia**, n (%) | 174 (9) | 77 (8) | 251 (9) |
| **Type 2 diabetes mellitus**, n (%) | 950 (49) | 481 (50) | 1,431 (49) |
| **Lipid Lowering Therapies** | | | |
| **No statin therapy** | **65 (3)** | **35 (4)** | **100 (3)** |
| **Any Statin Therapy, n (%)** | **1,875 (97)** | **934 (96)** | **2,809 (97)** |
| Low intensity statin therapy | 24 (1) | 11 (1) | 35 (1) |
| Moderate intensity statin therapy | 812 (42) | 384 (40) | 1,196 (41) |
| High intensity statin therapy | 1,039 (54) | 539 (56) | 1,578 (54) |
| **Cholesterol-absorption inhibitor (e.g. ezetimibe)** | 497 (26) | 254 (26) | 751 (26) |
| **Baseline Lipid Levels** | | | |
| **Baseline LDL-C**, mean (SD) mg/dL | 95.0 (38.8) | 98.3 (39.2) | 96.1 (38.9) |

CORALreef Lipids

# Enlicitide was well tolerated with safety profile similar to placebo and high patient adherence

| | Enlicitide | Placebo |
|---|---|---|
| | N = 1,935<br>n (%) | N = 969<br>n (%) |
| ≥1 Adverse Events | 1,244 (64.3) | 602 (62.1) |
| Serious AEs | 191 (9.9) | 116 (12.0) |
| Moderate or severe AEs | 649 (33.5) | 314 (32.4) |
| Discontinued Intervention due to an AE | 60 (3.1) | 40 (4.1) |
| Discontinued due to drug-related AEs | 30 (1.6) | 19 (2.0) |
| Discontinued due to a serious drug-related AE | 1 (0.1) | 0 (0.0) |
| Deaths | 13 (0.7) | 7 (0.7) |

Over 52-week study

- **97% adherence with study intervention** across treatment groups
- **≥97% adherence with fasting instructions** across treatment groups

CORALreef Lipids

The table summarizes on-treatment AEs that occurred after the first dose and up to either the last dose and up to either the last dose before rollover into the open label extension or up to the 8-week follow-up period after the last dose. Three deaths occurred >8 weeks after the last dose and are not included here

# No meaningful differences in proportion of participants with AEs between groups

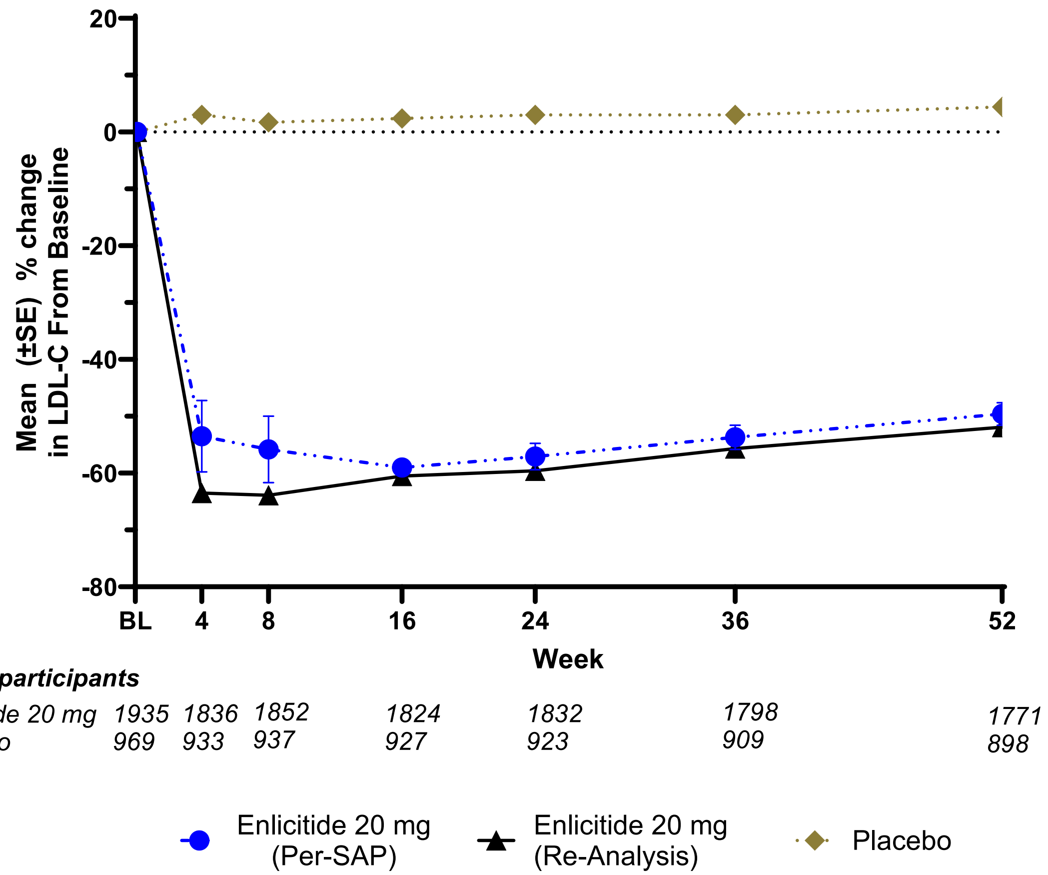
AEs reported in ≥10% of patients

| | Enlicitide | Placebo |
|---|---|---|
| | **N = 1,935<br>n (%)** | **N = 969<br>n (%)** |
| Gastrointestinal disorder | 296 (15.3) | 154 (15.9) |
| Infections and Infestations | 467 (24.1) | 246 (25.4) |
| Injury, poisoning and procedural complications | 218 (11.3) | 94 (9.7) |
| Metabolism and nutrition disorders | 194 (10.0) | 94 (9.7) |
| Musculoskeletal and connective tissue disorders | 252 (13.0) | 132 (13.6) |
| Nervous system disorders | 209 (10.8) | 91 (9.4) |
| Gastrointestinal disorder | 296 (15.3) | 154 (15.9) |

CORALreef Lipids

# Enlicitide is the first oral PCSK9i designed to have antibody-like efficacy: statistically significant, clinically profound and durable reductions in LDL-C

## **Primary endpoint**: Mean percent change in LDL-C




*No. of participants*

| | BL | 4 | 8 | 16 | 24 | 36 | 52 |
|---|---|---|---|---|---|---|---|
| *Enlicitide 20 mg* | *1935* | *1836* | *1852* | *1824* | *1832* | *1798* | *1771* |
| *Placebo* | *969* | *933* | *937* | *927* | *923* | *909* | *898* |

**55.8%**

LDL-C reduction from baseline at Week 24

**59.7%**

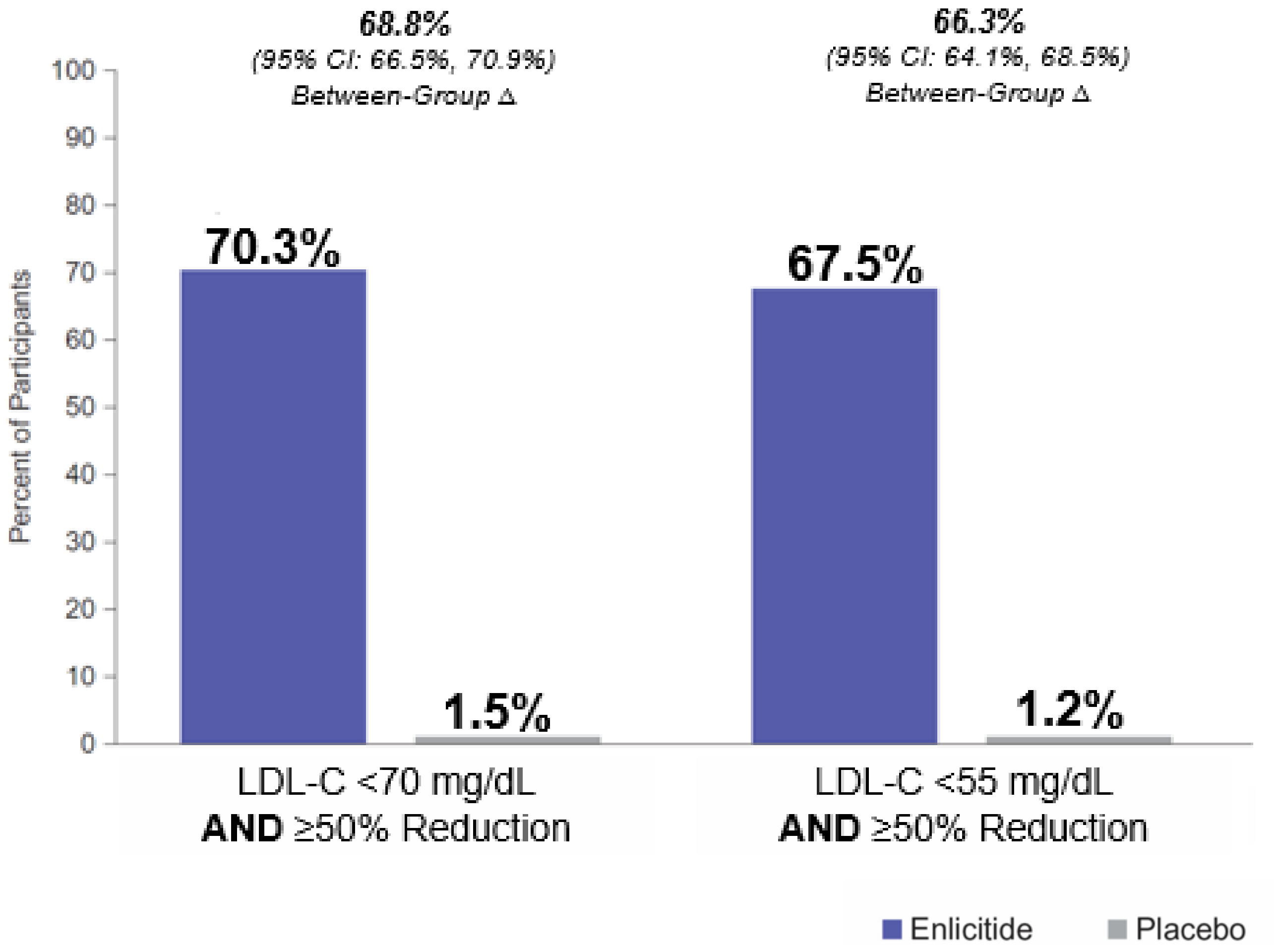
LDL-C reduction from baseline at Week 24 (post-hoc[1])

LDL-C reduction **seen early** and **durable** over 52 weeks[2]

CORALreef Lipids

1. Post-hoc reanalysis conducted to remove imputed biologically impossible baseline values. Refer to Appendix for additional detail 2. Secondary endpoint LDL-C mean percent change from baseline at Week 52

# Majority of participants achieved rigorous LDL-C goals




**Most participants reached study defined LDL-C goals,** an important factor in lowering cardiovascular disease risk

**Over two-thirds** of participants achieved rigorous prespecified goal **requiring both** LDL-C <55 mg/dL **and** ≥50% reduction from baseline[1]

CORALreef Lipids

1. Non-multiplicity controlled secondary endpoint

# Enlicitide is the first oral PCSK9 designed to have antibody-like efficacy: robust effect size on other important atherogenic lipoprotein biomarkers

Secondary Endpoints

| ApoB reduction | Non-HDL-C reduction | Lp(a) reduction |
|---|---|---|
| -50.3% | -53.4% | -28.2% |

CORALreef Lipids

# CORALreef HeFH evaluated efficacy of enlicitide in adults with heterozygous familial hypercholesterolemia

**Inclusion Criteria:**

HeFH with history of major ASCVD event & LDL-C ≥55 mg/dL
or
HeFH with no history of major ASCVD event & LDL-C ≥70 mg/dL

**Treatment period: 52 Weeks**

R: 2:1

**Total randomized: 303**

- Enlicitide 20 mg oral daily (on top of statins ± other lipid lowering therapies)
- Placebo (on top of statins ± other lipid lowering therapies)

**8 weeks**

Follow-up or open-label extension

**Primary Endpoint**

- LDL-C mean percent change from baseline at Week 24
- Proportion of participants with AEs; proportion of participants with discontinuations due to AEs

**Secondary Endpoints**

**Multiplicity-controlled secondary endpoints**

- LDL-C mean percent change from baseline at Week 52
- non-HDL-C mean percent change from baseline at Week 24
- ApoB mean percent change from baseline at Week 24
- Lp(a) median percent change from baseline at Week 24

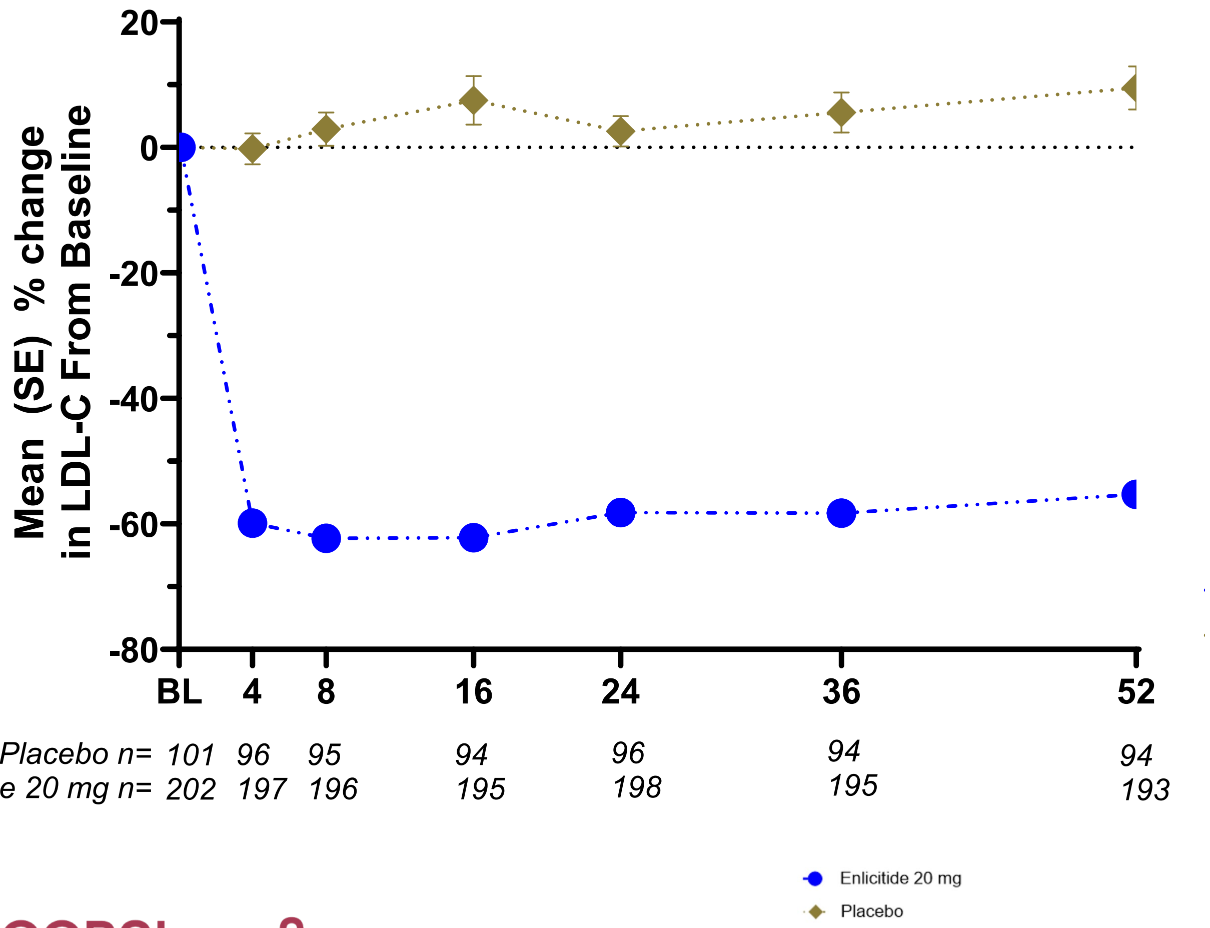
**Other secondary endpoints**

- Proportion of participants with LDL-C <70 mg/dL and ≥50% reduction from baseline at Week 24
- Proportion of participants with LDL-C <55 mg/dL and ≥50% reduction from baseline at Week 24

CORALreef HeFH

# Statistically significant, clinically profound and durable reductions in LDL-C in CORALreef HeFH

**Mean percent change in LDL-C over time**




# 59.4%

LDL-C reduction from baseline at Week 24

LDL-C reduction **seen early** and **durable** over 52 weeks[1]

CORALreef HeFH

1. Secondary endpoint LDL-C mean percent change from baseline at Week 52

# Consistently robust LDL-C lowering across enlicitide development program

| | Enlicitide Phase 2 Study | CORALreef Lipids | CORALreef HeFH | CORALreef AddOn |
|---|---|---|---|---|
| % LDL-C reduction from baseline (enlicitide) | **-57.0** (18mg)<br>**-59.7** (30mg) | **-57.1**[2], **-59.6**[3] | **-58.2** | Statistically significant and clinically meaningful[4] |
| Placebo-adjusted % LDL-C reduction from baseline[1] (enlicitide versus placebo) | **-58.3** (18 mg)<br>**-60.9** (30mg) | **-55.8**[2], **-59.7**[3] | **-59.4** | n/a<br>(no placebo group) |

1. Difference LS Mean 2. Primary analysis 3. Post-hoc reanalysis conducted to remove imputed biologically impossible baseline values 4. Topline results announced June 2025, enlicitide successfully met primary and all key secondary endpoints, demonstrating statistically significant and clinically meaningful greater reductions in LDL-C compared to other oral non-statin therapies

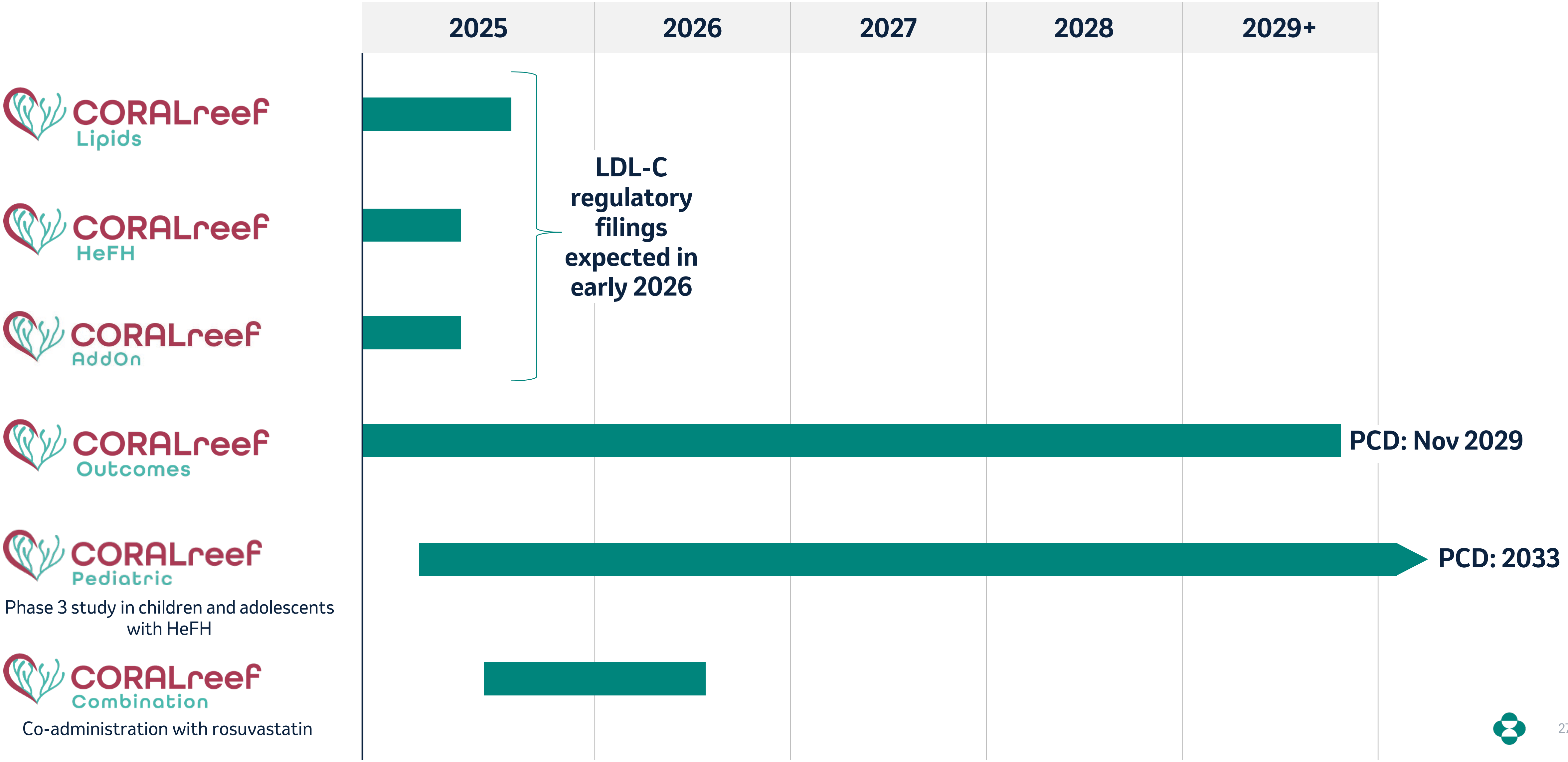
# Enlicitide demonstrated consistently robust efficacy for other atherogenic predictors of cardiovascular disease

| Secondary Endpoints (placebo-adjusted) | Enlicitide Phase 2 Study | CORALreef Lipids | CORALreef HeFH |
|---|---|---|---|
| % reduction in ApoB | **-47.8** (18mg)<br>**-51.8** (30mg) | **-50.3** | **-49.1** |
| % reduction in non-HDL | **-53.2** (18mg)<br>**-55.8** (30mg) | **-53.4** | **-53.0** |
| % reduction in Lp(a) | **-21.7** (18mg)<br>**-23.7** (30mg) | **-28.2** | **-27.5** |

# Phase 3 results support ambition to bring first oral PCSK9i designed to have antibody-like efficacy to broad population globally

- **Profound LDL-C reduction** exceeding other oral lipid lowering therapies used as an add-on to statins
- **Ease of use**, with placebo-like safety and tolerability as well as no adverse DDIs
- **Robust LDL-C goal attainment** in broad range of primary and secondary prevention patients
- Data supportive of goal in CORALreef Outcomes to **reduce the risk for stroke, MI and CV death**
- **Broad** atherogenic particle reduction
- Potential **anchor for combinations**

# Phase 3 CORALreef Lipids and HeFH studies support filing in early 2026



# Enlicitide is designed to help address the CV epidemic and anchor future novel combinations

**Now**

## Establish Foundation

Pending approval, aim to establish enlicitide as the **preferred** add-on lipid lowering therapy based on compelling CORALreef Lipids, HeFH and AddOn results

**Near**

## Elevate Outcomes Expectations

Goal of demonstrating **robust risk reduction** through ongoing **CVOT study**

Enable rapid, intensive escalation of lipid lowering therapy through **FDCs with enlicitide anchor**

**Next**

## Further Transform Treatment

Expand into other **atherogenic mechanisms** and **overlapping risks**

Pipeline includes:

- Lp(a) inhibition: MK-7262
- Multiple early programs in inflammation

# Addressing the Silent CV Epidemic

Jannie Oosthuizen
President, U.S. Human Health

MERCK

# Improving LDL-C control can materially reduce cardiovascular events and positively impact the global CV epidemic

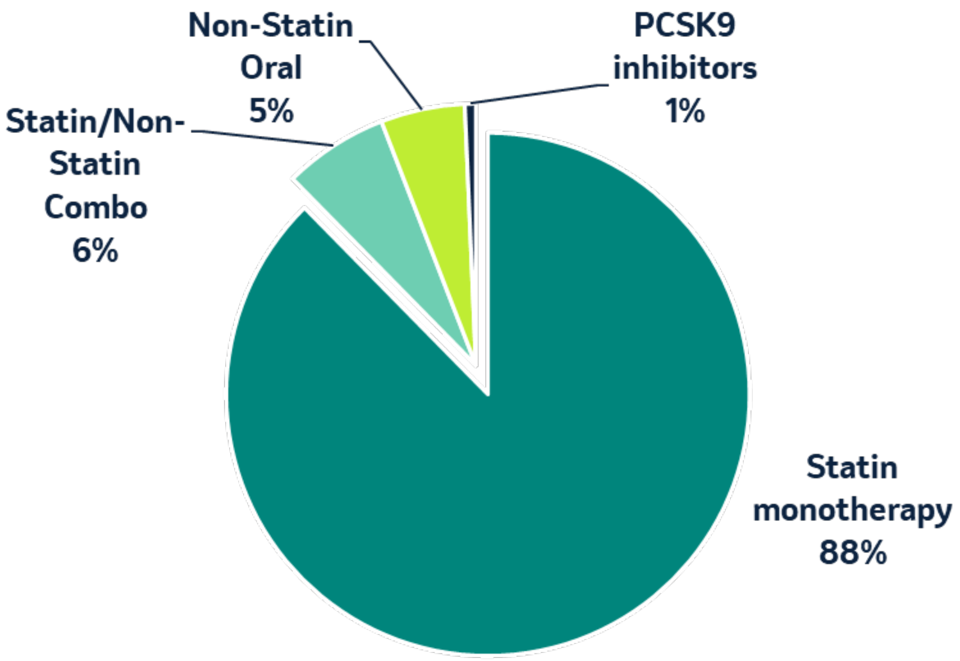
## Urgent need to address CVD

**#1**
**cause of mortality** globally

**85%**
of CV disease-related **deaths** each year are **due to ASCVD**

**Substantial**
economic burden

## 2024 Global Treated Patients[1]




**Vast majority** of treated patients are only on statin monotherapy

1. IQVIA

# Initial launch of enlicitide focused on patients on lipid lowering therapy who require additional LDL-C lowering to achieve their goal

## Current State

**Low patient awareness** of "asymptomatic" disease

Guidelines and quality measures vary with **suboptimal implementation**

**Limited use** of PCSK9 class of therapies to date

## Early Launch Focus

**>80M**
Estimated treated, not-at-goal ASCVD and at-risk patients in the U.S., Europe, and Japan[1]

**~ 45%**
ASCVD (Secondary Prevention)

**~ 55%**
At Risk for ASCVD (Primary Prevention)

Includes ~4M with HeFH

1. Excludes low risk

# Enlicitide is an investigational oral PCSK9i designed to have antibody-like efficacy and help address the silent CV epidemic

| | |
|---|---|
| **Profound efficacy** | A **new bar** for oral add-on lipid-lowering therapy |
| **Robust LDL-C goal attainment** | Potential to achieve **more rigorous guideline goals** |
| **Ease of use** | Expected to enable **broad real-world uptake** |
| **Potential for broad access** | Commitment to working with payers to **reduce barriers** and **support patient access** |
| **Expanded reach into primary care** | Designed to help **address the CV Epidemic** |

# Strong growth anticipated from cardiometabolic and respiratory programs well into next decade

**~$15B**

Commercial opportunity by mid-2030s[1]

Enlicitide has **multibillion-dollar potential**

+

New **multibillion opportunity** with

Ohtuvayre™ (ensifentrine) Inhalation Suspension 3 mg/2.5 mL

**Early-phase programs and deep discovery research efforts**

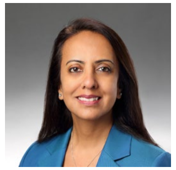
1. Non-risk adjusted

# Q&A

**Dr. Dean Li**
President, Merck Research Laboratories

**Dr. Joerg Koglin**
SVP, Head of General & Specialty Medicine, Global Clinical Development

**Jannie Oosthuizen**
President, U.S. Human Health



**Kshama Roberts**
SVP, Global Pharmaceuticals

**Peter Dannenbaum**
SVP, Investor Relations

# Appendix

# Effect of WINREVAIR on mortality and major morbidity outcomes in patients with pulmonary arterial hypertension: pooled analysis of the PULSAR, STELLAR, and ZENITH trial

| Pooled Analysis of the PULSAR, STELLAR, and ZENITH | | | |
|---|---|---|---|
| | **WINREVAIR**<br>**n=323** | **Placebo**<br>**n=278** | **HR (95% CI); P-value** |
| **Time to composite first major morbidity/mortality endpoint** | 25 (7.7) | 69 (24.8) | 0.25 (0.16, 0.40); <0.0001 |
| **Components of composite endpoint** | | | |
| All cause death | 14 (4.3) | 21 (7.6) | |
| Lung transplantation | 2 (0.6) | 6 (2.2) | |
| PAH-worsening related hospitalization of ≥24 hours | 10 (3.1) | 59 (21.2) | |
| **Transplant-free survival** | 16 (5.0) | 27 (9.7) | 0.44 (0.24, 0.83); 0.005 |
| **Overall survival (all cause death)** | 14 (4.3) | 21 (7.6) | 0.49 (0.25, 0.98); 0.019 |

# Acronyms

**AE** = Adverse Event

**ApoB** = Apolipoprotein B

**ASCVD** = Atherosclerotic cardiovascular disease

**BMI** = Body mass index

**COPD** = Chronic obstructive pulmonary disease

**CV** = Cardiovascular

**DDI** = Drug-drug interactions

**FDC =** Fixed dose combination

**GLP-1** = Glucagon-like peptide-1

**HeFH** = Heterozygous familial hypercholesterolemia

**HDL-C** = High-density lipoprotein cholesterol

**LDL-C** = Low-density lipoprotein cholesterol

**LLT** = Lipid lowering therapy

**Lp(a)** = Lipoprotein a

**mAb** = Monoclonal antibody

**MACE** = Major adverse cardiovascular events

**PAH** = Pulmonary arterial hypertension

**PCSK9** = Proprotein convertase subtilisin/kexin type 9

**PDE3** = Phosphodiesterase 3

**PDE4** = Phosphodiesterase 4

# Dean Li, M.D., Ph.D.
## President, Merck Research Laboratories

Dr. Dean Li serves as executive vice president and president of Merck Research Laboratories (MRL). He leads the company's worldwide human vaccines and therapeutics research and development organization.

Since joining Merck in 2017, Dean has held leadership roles in the Translational Medicine and Discovery functions and was appointed to President, Merck Research Laboratories in January 2021. Prior to joining Merck, Dean held positions of increasing responsibility in translational medical research at the University of Utah. Most recently, he served as the H.A. & Edna Benning Professor of Medicine and Cardiology, chief scientific officer, associate vice president and vice dean at the University of Utah Health System. From 2015 to 2016, he also served as interim CEO of Associated Regional University Pathologists, one of the United States' largest clinical reference laboratories. During his tenure at the University of Utah, he co-founded several biotechnology companies based upon research conducted in his laboratory, including Recursion Pharmaceuticals, Hydra Biosciences and Navigen Pharmaceuticals.

Dean received his Bachelor's degree in Chemistry from the University of Chicago and his graduate and clinical training at Washington University School of Medicine in St. Louis. Dean is a board-certified cardiologist, a member of the American Society for Clinical Investigation and the Association of American Physicians.

# Joerg Koglin, M.D., Ph.D.

## Senior Vice President, Head of General & Specialty Medicine, Global Clinica

Dr. Joerg Koglin serves as senior vice president of Merck Research Laboratories (MRL) and head of General and Specialty Medicine Global Clinical Development.

Joerg earned his MD from the University of Heidelberg, Germany and his PhD from the University of Munich, Germany, and completed a postdoctoral fellowship at the Cardiovascular Biology Laboratory of the Harvard School of Public Health. He is board-certified in Internal Medicine and Cardiology. After more than 10 years as an academia-based physician with a junior faculty position at the Department of Cardiology, University of Munich, Germany, Joerg transitioned to corporate research and development, where he has worked for over 20 years. Since joining MRL in 2007 in the Late-Stage Global Clinical Development organization, Joerg has held various roles including Clinical Lead, Development Team Lead, Section Head, and Therapeutic Area Head for Cardiovascular, Respiratory and Metabolism.

In his current role within Global Clinical Development, Joerg oversees clinical development efforts across the General and Specialty Medicine therapeutic areas, which include Cardiovascular & Respiratory, Atherosclerosis and Metabolism, Immunology and Neurosciences, and Ophthalmology.

# Jannie Oosthuizen

## President, Human Health U.S.

Jannie Oosthuizen leads Merck Human Health U.S., which is Merck's largest business globally. He is responsible for P&L, strategy, customer engagement and commercialization in the U.S. for Merck's broad portfolio of human health medicines and vaccines.

Jannie joined the company in 2014 to lead the Human Health oncology business in Asia Pacific and Latin America, then led Merck's business in Japan from 2016 to 2020, and then led Global Marketing for Oncology. In each of these roles, Jannie successfully created and implemented new strategies and innovative commercial models that delivered strong, leveraged growth and established Merck as a leading business in those markets and therapeutic areas.

Jannie has deep experience in a broad range of global markets and therapeutic areas, and in building and leading high-performing teams. Prior to Merck, Jannie spent 20 years at Eli Lilly in a wide range of commercial and marketing roles with increasing responsibility. He began his career with Eli Lilly in 1993 in his home country of South Africa. Jannie is a pharmacist by training and graduated from North-West University in South Africa.

# Kshama Roberts

## Senior Vice President, Human Health U.S.

Kshama Roberts leads the global pharmaceuticals franchise, overseeing the Human Health commercial team responsible for the portfolio and late-stage pipeline across multiple therapeutic areas, including cardiometabolic, immunology, ophthalmology, and neuroscience. Kshama is responsible for developing strategy and commercial execution globally for a growing and diverse portfolio of important medicines, including WINREVAIR, a first-in-class therapy for pulmonary arterial hypertension.

Kshama joined Merck in 1994, with most of her experience in Human Health leading global brand marketing and commercialization teams across many of the company's top brands. Prior to her current role, Kshama served as the chief of staff for the chairman and CEO, working with the CEO and executive team to define and support the company's strategic and enterprise-wide operational priorities and driving execution excellence across the organization. Kshama also headed Merck's corporate strategy office, where she led the spinoff of Organon, including a portfolio of more than 60 medicines and products across three core franchises.

Kshama began her career at Merck's MSD U.K. subsidiary. Kshama graduated from the London School of Economics & Political Sciences with a Bachelor of Science in management sciences specializing in marketing and statistics.